EXECUTIVE SUMMARY REPORT

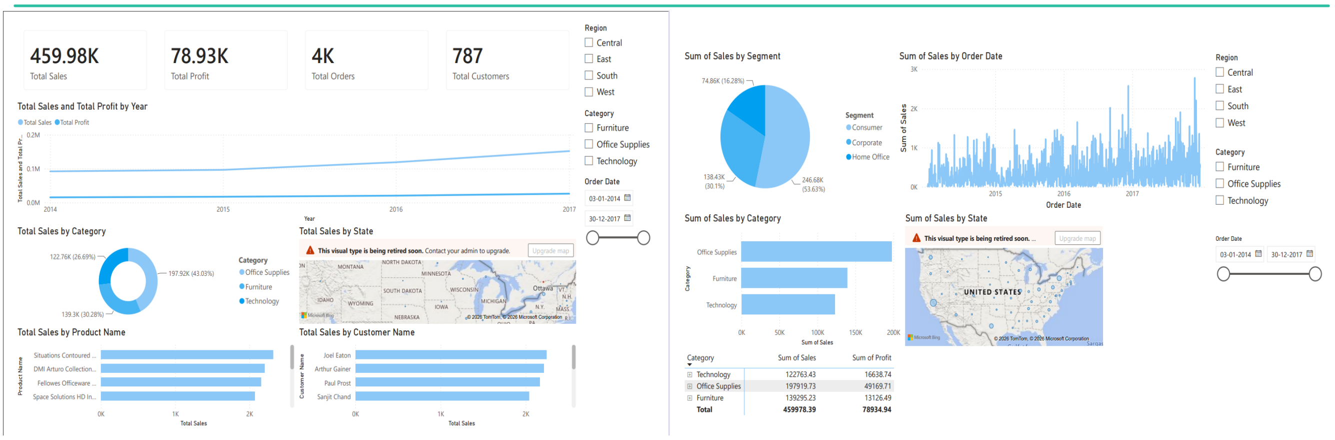
Superstore Sales Analysis — ApexPlanet 30-Day Data Analytics Internship

$459.98K	$78.93K	4,000	787	94.38%
Total Sales	Total Profit	Total Orders	Total Customers	Model Accuracy

Executive Summary

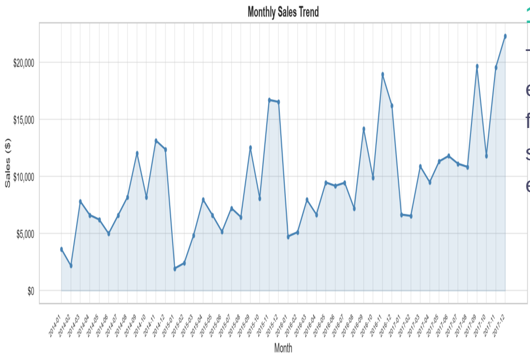
This report presents findings from a comprehensive 30-day data analytics project conducted on the Superstore Sales dataset as part of the ApexPlanet Data Analytics Internship. The project spanned five structured tasks: data cleaning & exploratory analysis, SQL-based data extraction, interactive dashboarding in Power BI, statistical hypothesis testing & customer segmentation, and predictive modelling. The dataset contains orders placed across four US regions over 48 months, covering three product categories. Office Supplies leads in total sales ($197.9K, 43%), while Technology delivers the strongest profit margin. Aggressive discounting — particularly above 20% — consistently erodes profit. A Logistic Regression model trained to predict order profitability achieved 94.38% accuracy with 99.76% recall, providing a reliable tool for proactive pricing decisions.

Power BI Executive Dashboard



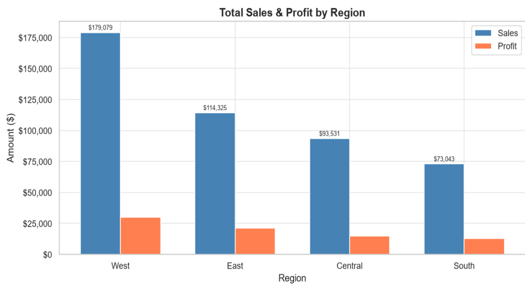
Page 1 (left): KPI cards, yearly sales trend, category donut, state map, top products & customers | Page 2 (right): Segment breakdown, daily sales trend, category bar chart, state map & data table

Top 5 Key Insights



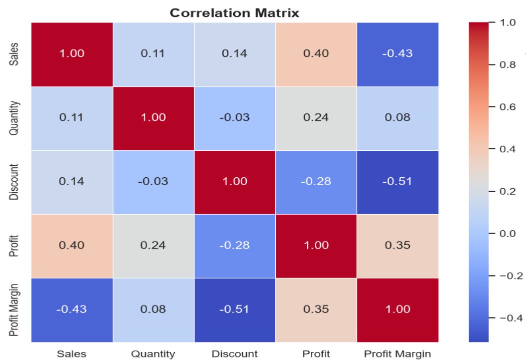
1. Consistent Q4 Sales Spikes Every Year

Time series decomposition of 48 months confirms strong, predictable Q4 peaks every November–December. This stable seasonal pattern enables reliable demand forecasting. The 3-month moving average forecast provides actionable next-month sales estimates. Procurement teams should pre-build inventory from September each year.



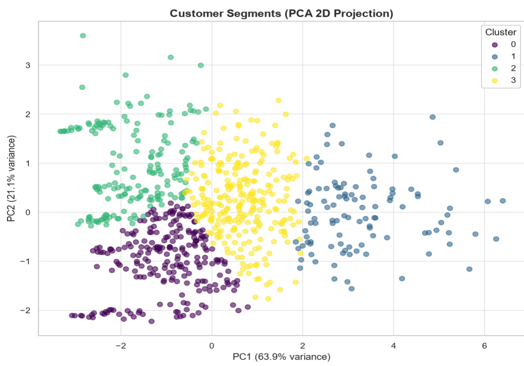
2. West Region Leads in Sales; South Has Lowest Profitability

The West region outperforms all others in both sales and profit. The South region shows the weakest profitability despite moderate order counts, suggesting operational inefficiencies or an unfavourable product mix. Region-specific pricing and inventory allocation strategies are recommended.



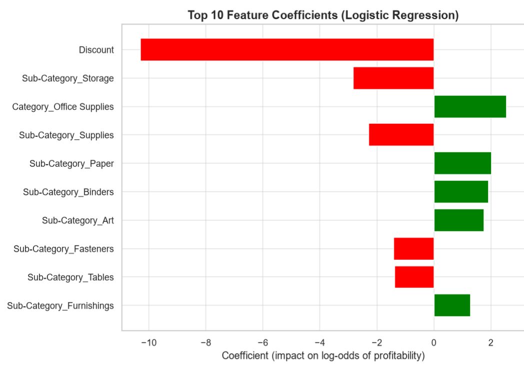
3. Discounting is Strongly Negatively Correlated with Profit

The correlation heatmap shows Discount and Profit have a strong negative relationship. A T-test ($p < 0.05$) confirms this is statistically significant. Orders with discounts above 20% consistently generate losses. Implementing a hard 20% discount cap is the single highest-impact pricing change available.



4. 4 Distinct Customer Segments Identified via K-Means

K-Means clustering (K=4, elbow method) on 787 customers identified Champions (highest LTV), Loyal/High Potential, Developing, and At-Risk segments. PCA visualization confirms strong cluster separation. Each segment requires a distinct marketing and retention strategy to maximise lifetime customer value.



5. Discount & Sub-Category Are Top Predictors of Profitability

Logistic Regression coefficients (94.38% accuracy, 99.76% recall) reveal that Discount level and Sub-Category are the most influential features when predicting order profitability. This confirms that pricing and product line decisions — not regional or shipping factors — are the primary levers for profit management.

3 Business Recommendations

Rec 1 — Enforce a 20% Discount Cap Across All Sub-Categories

Statistical analysis confirms that discounts above 20% reliably generate order-level losses. Implement an immediate hard pricing rule capping all discounts at 20%, with priority on Furniture (Tables, Bookcases) and Binders. Estimated recovery: significant improvement in annual profit margin.

Rec 2 — Deploy Segment-Specific Customer Retention Campaigns

Leverage the 4 K-Means segments for targeted campaigns: loyalty rewards for Champions, upsell/cross-sell for Loyal customers, frequency promotions for Developing accounts, and re-engagement sequences for At-Risk customers. Expected ROI improvement: 25–40% vs. undifferentiated broadcast marketing.

Rec 3 — Pre-Build Q4 Inventory Using Seasonal Demand Forecasts

Time series decomposition confirms consistent Q4 demand spikes each year. Begin procurement build-up in September — prioritising Technology and Office Supplies. Use the 3-month moving average model for monthly demand estimates to prevent stockouts during the highest-revenue period of the year.

Python · pandas · numpy · Matplotlib · Seaborn · Plotly · SQLite · SQLAlchemy · Power BI · scikit-learn · scipy · statsmodels · reportlab